Notes: Jäger, Schoefer, and Zweimüller (RES, 2023)
Marginal Jobs and Job Surplus: A Test of the Efficiency of Separations

Contents

1 Big picture

- **Question.** Are job separations bilaterally efficient, as in Coasean models of transferable surplus, or can labor-market frictions such as wage rigidity cause inefficient separations?
- **Core idea.** The paper designs a revealed-preference test of efficient separations. A temporary policy shock raises workers' outside option, destroys low-surplus jobs in the treatment group, and is then repealed. If separations are Coasean and efficient, the surviving treated jobs should be missing the low-surplus marginal matches and should be more resilient to future shocks.
- **Empirical setting.** The Austrian Regional Extended Benefit Program (REBP) increased unemployment-insurance potential benefit duration for older workers in treated regions from roughly 20–30 weeks to 209 weeks. The program started in 1988 and was sharply repealed in 1993.
- **Treatment variation.** Eligibility varied by region and cohort: workers aged 50 or older in REBP regions were eligible, while younger workers and workers in control regions were not. This supports a difference-in-differences design comparing older and younger cohorts across treated and control regions.
- **First empirical fact.** During REBP, treated eligible jobs separated substantially more often. The separation effect is about 11 percentage points, implying that roughly 18.5% of the control-group survivors are marginal jobs that would have been destroyed by the REBP shock.
- **Second empirical fact.** After REBP was repealed, the formerly treated survivors were *not* more resilient. Their post-repeal separation behavior is statistically indistinguishable from the former control-group survivors.
- **Coasean prediction.** If bargaining is bilaterally efficient, the marginal low-surplus matches should remain in the control group but not in the treated survivor group. Therefore, post-repeal shocks should first destroy control-group marginal jobs, producing lower separations among former treatment survivors.
- **Result.** The data reject this prediction. The authors show that simple Coasean benchmarks, mixed reshuffling models, and calibrated idiosyncratic-shock versions still predict substantial resilience that is absent in the data.
- **Mechanism.** A non-Coasean model with wage rigidity can explain the result. The initial UI extension selects jobs on worker surplus rather than joint surplus, so the destroyed matches are not necessarily the lowest joint-surplus matches. Post-repeal, former treatment survivors need not be especially robust to firm-side shocks.
- **Wage rigidity evidence.** The evidence is suggestive rather than causal, but the non-resilience pattern is stronger in firms or cells with less wage dispersion, which proxy for greater wage rigidity.
- **Economic takeaway.** Separations are not always efficient bilateral bargains. Wage-setting frictions can make matches dissolve even when joint surplus remains positive, so unemployment-insurance policy and labor-market shocks can have allocative consequences beyond voluntary surplus-sharing.

2 Incremental contribution of the paper

2.1 Contribution relative to Coasean labor-market theory

- The dominant theoretical view assumes that workers and firms can bargain over wages or transfers until all positive joint-surplus matches survive.
- Prior evidence on wage rigidity does not by itself reject Coasean efficiency, because even fixed observed wages can be part of a deeper efficient contract with present-value transfers, implicit payments, or continuation arrangements.
- The paper contributes a direct revealed-preference test: it does not need to observe surplus levels. Instead, it uses the *order* in which jobs should separate if low-surplus matches have already been extracted.

- The test is sharper than standard wage-rigidity tests because it studies whether the composition of surviving jobs has changed in the way a Coasean model predicts.

Interpretation: The incremental theoretical contribution is to translate an abstract efficiency assumption into an empirical prediction about post-treatment resilience. The paper asks not merely whether wages are rigid, but whether rigidity is allocative for separations.

2.2 Contribution relative to unemployment-insurance research

- Existing Austrian REBP work had documented effects on unemployment entry, unemployment duration, job finding, spillovers, disability insurance, and early retirement.
- This paper studies a different object: whether the jobs destroyed by a UI-induced outside-option shock were low joint-surplus jobs.
- The innovation is exploiting the repeal. The initial reform identifies a mass of marginal matches; the repeal tests whether the survivors have become more resilient.
- The paper therefore moves from the reduced-form effect of UI on separations to the welfare-relevant efficiency property of the separations induced by UI.

Interpretation: The paper is not just another UI treatment-effect study. It uses UI policy as an experimental instrument for extracting marginal jobs and testing the organization of labor-market bargaining.

2.3 Contribution relative to wage-rigidity and rent-sharing evidence

- Many papers document wage stickiness, fairness constraints, rent sharing, or monopsony-like wage setting.
- A common objection is that wage rigidity need not matter for employment if parties can renegotiate or use implicit transfers.
- This paper provides evidence that wage rigidity can matter for *separations*, because the observed non-resilience is difficult to rationalize under efficient bargaining but natural under rigid wage differentiation.
- The wage-rigidity heterogeneity exercise links the main design to firm-level wage dispersion measures, showing that the initial REBP separation effect and the rejection of the Coasean benchmark are most pronounced where wages appear more rigid.

Interpretation: The incremental empirical contribution is to connect wage rigidity to inefficient match destruction rather than only to wage levels.

2.4 Contribution relative to empirical industrial organization of matches

- The design treats jobs as match-specific surplus objects and uses administrative data to follow the same matches from before the policy, through the policy, and after repeal.
- The paper turns a one-time shock into a dynamic composition test: the initial treatment changes the set of surviving jobs; post-repeal outcomes reveal whether the removed matches were the lowest joint-surplus matches.
- The approach does not require estimating match surplus directly in the main data, which would be hard because worker and firm outside values are unobserved.

Interpretation: The methodological contribution is the "missing mass of marginal matches" design: remove marginal matches in one group, preserve them in another, and then test whether subsequent separations occur in the predicted order.

3 Institutional context, data, and measurement

3.1 Austrian UI system and REBP policy

- REBP was a regional extended-benefit program introduced in 1988.
- The program sharply increased potential benefit duration for eligible workers from roughly 20–30 weeks to 209 weeks.
- The policy affected workers aged 50 or older who lived in REBP districts and had sufficient employment history.
- The program was repealed in 1993, which is essential for the paper because the repeal removes the outside-option shock while leaving behind a changed composition of survivors.
- The UI replacement rate was roughly 40–48% for many workers, but the duration extension was large. The authors ballpark the present value of the extension as about 71% of annual salary.

Interpretation: REBP is useful because it is large, temporary, and targeted. It creates an outside-option shock that can destroy marginal jobs and then disappears, permitting a post-repeal resilience test.

3.2 Eligibility dimensions

- **Region:** workers in REBP regions were potentially eligible; workers in control regions were not.
- **Age/cohort:** workers had to be at least age 50 during the program to be covered.
- **Employment history:** workers needed sufficient employment experience.
- **Timing:** the program started in 1988 and was repealed in 1993.

Interpretation: The two-dimensional eligibility rule lets the authors compare older versus younger cohorts within regions and treated versus control regions within cohorts. This absorbs many region-level and cohort-level confounds.

3.3 Administrative data

- The main data are Austrian Social Security Database records matched to employer–employee information.
- The data cover private-sector, dependently employed, non-tenured public-sector workers.
- The core sample consists of workers employed at the onset of the reform in 1988q2.
- The sample follows whether those workers remain with the same employer, separate, enter non-employment, receive UI benefits, or remain continuously employed.
- The authors drop women because their experience data are less reliable in the period and because they could retire earlier.
- They also drop steel-sector workers and partially treated regions to avoid direct targeting or complicated treatment timing.

Interpretation: The data are match-level administrative data. This is crucial: the paper follows the same worker–employer match before, during, and after policy treatment.

3.4 Main outcomes

- **Separation:** an indicator that the worker is no longer employed by the 1988q2 employer by the endpoint.
- **Separation into non-employment:** separation followed by no employment with another employer.
- **Non-employment quarters:** time spent out of employment.
- **Unemployment benefit quarters:** time receiving UI or unemployment assistance.
- **Continuous employment quarters:** time continuously employed with the initial employer.

Interpretation: The separation outcome captures match destruction. The additional outcomes show that REBP separations are not mostly job-to-job reallocations; they often move workers into non-employment or benefit receipt.

3.5 Treatment and control groups

- The main treated cohort consists of eligible older workers in treatment regions.
- The ineligible cohort consists of younger workers who were not covered because they turned 50 only after the program ended.
- The control region contains analogous older and younger cohorts.
- Cohorts born before 1933 are excluded in key regressions because most had reached retirement age by 1993.
- Cohorts born after 1948 are excluded because they were still far below age 50 when the program ended.

Interpretation: The design is a difference-in-differences in region and birth cohort. The eligible-control comparison is not only across regions and not only across ages; it is their interaction.

4 Theory and models

4.1 Job surplus under bilaterally efficient bargaining

A worker–firm match has worker surplus and firm surplus:

$$S^W(w, V^W) = V^W_{In} + w - V^W_{Out} \geq 0, \tag{1}$$
$$S^F(w, V^F) = V^F_{In} - w - V^F_{Out} \geq 0. \tag{2}$$

Here w is the wage or transfer, V^W_{In} and V^F_{In} are within-match values, and V^W_{Out} and V^F_{Out} are outside values.

- If both surpluses are non-negative, the job is feasible.
- If worker surplus is negative and firm surplus is non-negative, the match ends as a quit.
- If firm surplus is negative and worker surplus is non-negative, the match ends as a layoff.
- If both are negative, the match ends as a mutual separation.

Interpretation: Under Coasean bargaining, the wage can transfer surplus between worker and firm. Therefore, the relevant viability condition collapses to joint surplus: a match with positive joint surplus should survive because the wage can be adjusted to satisfy both parties.

4.2 Joint surplus and efficient separations

Define the gross joint surplus of a match as

$$\widetilde{S}(V) = \left(V_{In}^{W} - V_{Out}^{W}\right) + \left(V_{In}^{F} - V_{Out}^{F}\right). \tag{3}$$

Under the Coasean view, separations occur if and only if

$$\widetilde{S}(V) < 0. \tag{4}$$

A wage can reallocate surplus but cannot change the total surplus.

Interpretation: The Coasean model predicts a pecking order of separation: low joint-surplus jobs separate before high joint-surplus jobs. This pecking order is what the empirical design tests.

4.3 Effect of REBP in the Coasean model

REBP raises the worker's non-employment outside option. In the authors' notation, this lowers worker surplus by a value ε_b^W and therefore lowers joint surplus by the same amount:

$$\widetilde{S}_{REBP}(V) = \widetilde{S}(V) - \varepsilon_b^W. \tag{5}$$

During the policy, jobs with

$$0 \leq \widetilde{S}(V) < \varepsilon_b^W \tag{6}$$

are destroyed in the treated group but remain in the control group.

Interpretation: These are the marginal jobs. Their removal is the source of the post-repeal prediction. After the policy ends, the former treatment group lacks low-surplus matches.

4.4 Coasean resilience benchmark

Let δ^1 be the during-REBP separation rate in the treated group and δ^0 the during-REBP separation rate in the control group. Let Δ^0 be the post-repeal separation rate among control survivors, and Δ^1 the Coasean-predicted post-repeal separation rate among treated survivors. With no post-repeal idiosyncratic shocks, the benchmark is

$$\Delta^1(\Delta^0, \delta^0, \delta^1) = \max\left\{0, \frac{1-\delta^0}{1-\delta^1}\Delta^0 - \frac{\delta^1-\delta^0}{1-\delta^1}\right\}. \tag{7}$$

The kink is at

$$\Delta^0 = \frac{\delta^1 - \delta^0}{1-\delta^0}. \tag{8}$$

Interpretation: The control group must first lose its preserved marginal jobs before the treatment group begins to separate. If the initial treatment effect is large, the kink is large and the model predicts substantial resilience.

4.5 Non-Coasean wage rigidity model

If wages are fixed or hard to differentiate across similar workers, worker and firm surpluses cannot be collapsed into one joint-surplus object. Separations occur when either side's participation constraint fails at the given wage:

$$\widetilde{d}(w, V; \varepsilon^W, \varepsilon^F) = \int \mathbf{1}\{\widetilde{S}^W(w', V^{W'}) < \varepsilon^W \ \vee \ \widetilde{S}^F(w', V^{F'}) < \varepsilon^F\} k((w', V')|(w, V))\, d(w', V'). \tag{9}$$

- REBP shifts worker surplus down, so it can select jobs with low worker surplus.
- These destroyed jobs need not be low in firm surplus or low in total joint surplus.
- After repeal, former treated survivors may be resilient to worker-surplus shocks but not to firm-surplus shocks.
- Negative demand shocks often operate on the firm-surplus margin, so non-resilience is natural under wage rigidity.

Interpretation: The non-Coasean model separates the worker-side and firm-side margins. This is why extracting low worker-surplus jobs does not necessarily create resilience to all future shocks.

4.6 Mixed Coasean reshuffling model

To assess whether idiosyncratic shocks can restore the Coasean prediction, the paper estimates a mixed model. For cohort c and labor-market cell i,

$$\Delta^1_{ci} = \kappa \Delta^0_{ci} + (1-\kappa) \sum_c \iota_c \max\left\{0, \frac{1-\delta^0_c}{1-\delta^1_c}\Delta^0_{ci} - \frac{\delta^1_c - \delta^0_c}{1-\delta^1_c}\right\} + \nu_{ci}. \tag{10}$$

- κ is the share of cells that would need full post-repeal surplus reshuffling.
- $1-\kappa$ is the share following the no-idiosyncratic-shock Coasean benchmark.
- Estimates put κ near one.

Interpretation: The Coasean model can rationalize the data only if almost all relevant cells experience essentially complete and rapid reshuffling of job surplus immediately after repeal, an assumption the authors argue is implausibly strong.

5 Empirical design and identification

5.1 Main difference-in-differences regression

The main DiD regression is

$$D_{rci} = \alpha + \beta\, REBPRegion_r + \gamma\, TreatedCohort_c + \mu\, (REBPRegion_r \times TreatedCohort_c) + \chi_{rci}. \tag{11}$$

- r indexes region.
- c indexes birth cohort.
- i indexes worker or job.
- D_{rci} is a separation or non-employment outcome.
- μ is the treatment effect of REBP eligibility.

Interpretation: The interaction of REBP region and treated cohort is the policy variation. Region and cohort main effects absorb baseline differences across places and age groups.

5.2 First stage of the revealed-preference test: marginal-job extraction

- The first empirical step documents that REBP raises separations during 1988–93.
- The estimated separation effect is 11.0 percentage points.
- The treatment group separation rate is 51.7%; the counterfactual control rate is 40.8%.
- Among control survivors, this implies that 18.5% are jobs that would have separated if exposed to REBP.

Interpretation: This step verifies that the policy extracted a sizable mass of marginal matches. Without this first-stage extraction, the post-repeal resilience test would have little power.

5.3 Second stage: post-repeal resilience

- The second step restricts the sample to jobs that existed at the onset of REBP and survived until repeal.
- The paper then compares post-repeal separations from 1994 to 1996 between former treatment and former control survivors.
- Under Coasean bargaining, former treatment survivors should be more resilient because low joint-surplus matches were already removed.
- In the data, the difference is essentially zero: 0.8 percentage points with standard error 1.0 percentage point.

Interpretation: The absence of post-repeal resilience is the central empirical fact. It says that the jobs removed during REBP were not simply the lowest joint-surplus matches.

5.4 Robustness to negative labor-demand shocks

- If the Coasean prediction is right, the former treatment group should be insulated from negative shocks because its marginal jobs are missing.
- The authors examine industry-growth and establishment-growth shocks.
- Former treatment survivors do not display less separation in shrinking industries or shrinking establishments.
- The slopes of separation with respect to establishment employment growth are similar across treatment and control groups.

Interpretation: The non-resilience result is not merely about average separations. It also holds when shocks are explicitly negative, exactly when the Coasean pecking-order prediction should have bite.

5.5 Alternative Coasean rationalizations

- **Idiosyncratic shocks:** random shocks could reshuffle surplus after repeal, refilling the treatment group's missing marginal matches.
- **Large shocks:** very large idiosyncratic shocks could swamp initial surplus composition.

- **Heterogeneous valuation of REBP:** the policy might remove high-surplus workers if they value the UI extension unusually much.
- **Homogeneity:** if all matches have nearly identical surplus, the treatment would not create a persistent compositional difference.
- **Large firms and substitution:** firms might reshuffle workers internally in ways that erase match-specific surplus differences.

Interpretation: The paper systematically considers these concerns. The authors conclude that the Coasean view can fit the data only with strong assumptions, especially near-complete post-repeal reshuffling.

5.6 Wage rigidity evidence

- The wage-rigidity proxies are based on within-firm wage dispersion and residual wage dispersion.
- Lower dispersion is interpreted as greater rigidity or less wage differentiation across similar workers.
- Initial REBP separations are larger in more rigid cells.
- The post-repeal data still show little resilience, and the gap with the Coasean benchmark is strongest where rigidity is high.

Interpretation: The wage-rigidity evidence is suggestive because rigidity measures are not randomly assigned. But it aligns with the non-Coasean mechanism and helps explain why efficient renegotiation fails.

6 Tables and figures

6.1 Figure 1 and Table 1: policy variation and sample balance

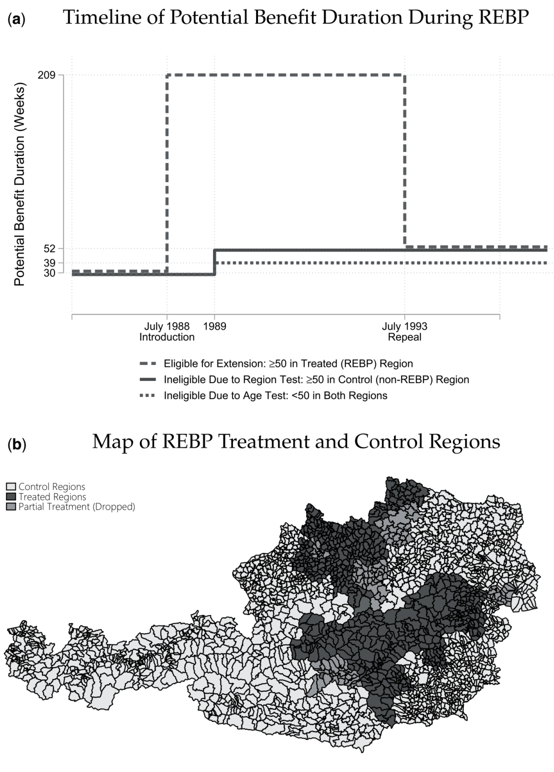
Read: Figure 1:

- Panel (a) shows the REBP treatment: potential benefit duration rises sharply to 209 weeks for eligible workers aged 50 or older in treated regions beginning in 1988.
- Control-region older workers receive only the nationwide increase to roughly 52 weeks.
- Younger workers in both regions remain ineligible for the large extension.
- Panel (b) maps treatment and control regions in Austria, with partially treated regions excluded.

Read: Table 1:

- Eligible and ineligible cohorts have similar annual earnings across treated and control regions.
- Eligible treated-region workers are older by construction and have about 22.1 years of experience and 11.2 years of tenure.
- The control region is larger: 198,124 eligible-cohort observations and 116,852 ineligible-cohort observations.
- White-collar shares differ by region, which motivates cohort-by-region comparisons rather than simple cross-sectional comparisons.

Economic message: The design has a large, sharp policy treatment and rich administrative data. The comparison must account for region and cohort differences, which is why the DiD structure is central.



(a) REBP timeline and treatment regions

TABLE 1
Summary statistics

	Treatment region		Control region	
	Eligible cohort (1)	Ineligible cohort (2)	Eligible cohort (3)	Ineligible coho (4)
Birth year	1938.636	1945.704	1938.749	1945.692
	(2.800)	(1.526)	(2.802)	(1.544)
Experience	22.138	20.536	20.988	19.150
	(5.408)	(5.509)	(6.006)	(6.100)
Tenure	11.168	9.630	10.075	8.719
	(5.885)	(6.027)	(5.941)	(5.932)
Annual earnings (1,000 EUR)	36.332	35.747	36.466	35.908
	(10.002)	(10.103)	(10.787)	(11.025)
White collar	0.378	0.401	0.470	0.483
	(0.485)	(0.490)	(0.499)	(0.500)
Observations	52,294	29,059	198,124	116,852

Notes: This table reports summary statistics—means and standard deviations (in parentheses)—for our sample of worke employed at the onset of the reform (1988q2). Columns (1) and (2) do so for the treatment regions and Columns (3) a (4) for the control regions, described in Section 2 and outlined in Figure 1(b). Columns (1) and (3) report on the eligib

(b) Summary statistics

6.2 Figure 2 and Figure 3: theoretical test of Coasean separations

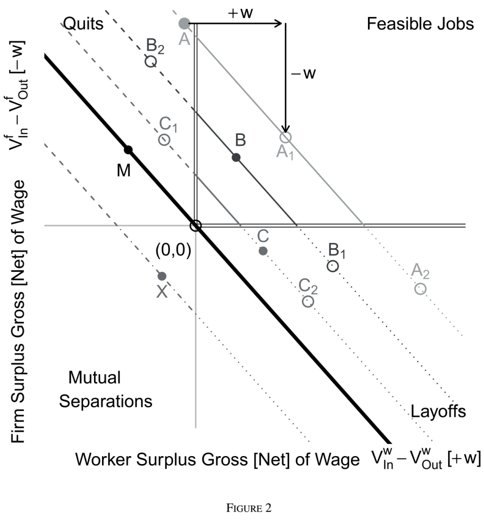
Read: Figure 2:

- The axes are worker surplus and firm surplus.
- Wages move matches along iso-joint-surplus lines.
- A match with positive joint surplus can be moved into the feasible region if transfers are flexible.
- A match with negative joint surplus cannot be saved by wage adjustment.

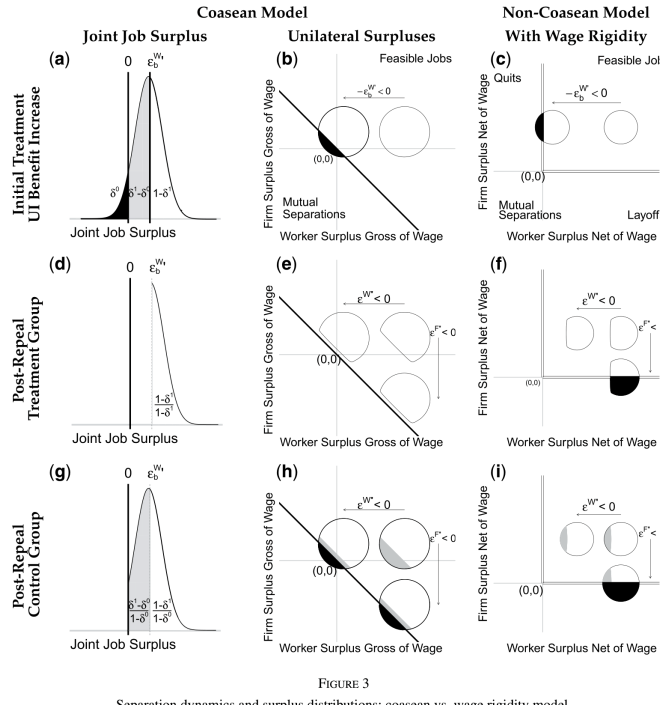
Read: Figure 3:

- The left column shows the Coasean joint-surplus representation.
- REBP removes low joint-surplus marginal jobs in the treated group.
- After repeal, the treatment survivor distribution should lack these low-surplus jobs.
- The right column shows the wage-rigidity alternative: REBP selects jobs with low worker surplus, not necessarily low joint surplus.

Economic message: The figures clarify the central contrast. Coasean bargaining predicts post-repeal resilience because the lowest joint-surplus jobs are gone; wage rigidity breaks this prediction because separations depend on unilateral participation constraints.



(a) Case studies of jobs



(b) Coasean vs. wage-rigidity model

6.3 Figure 4 and Table 2: the initial REBP separation effect

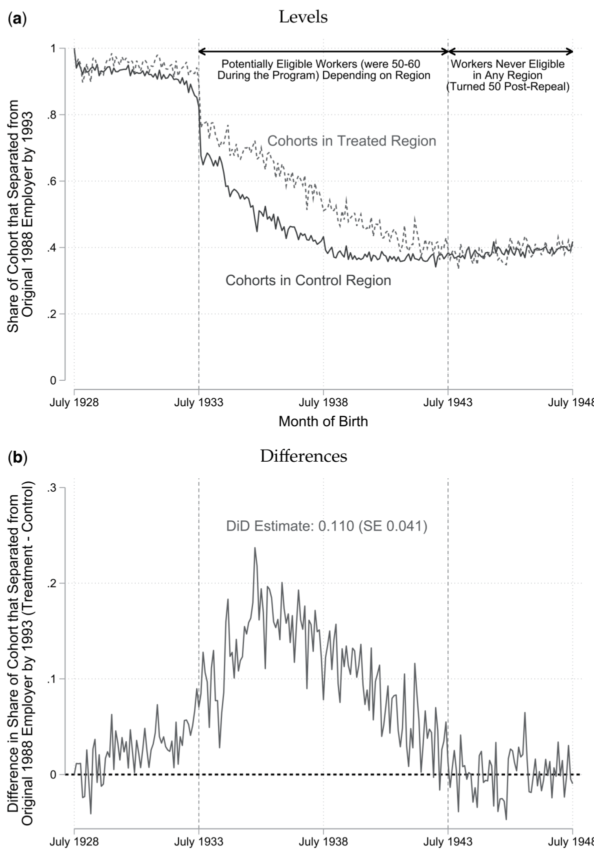
Read: Figure 4:

- Panel (a) shows that treated-region eligible cohorts separate more from their original 1988 employer by 1993.
- The separation gap opens for cohorts who were age-eligible during the program.
- Ineligible younger cohorts show no systematic treated-control gap, supporting the parallel-trends logic.
- Panel (b) reports a DiD estimate of 0.110 with standard error 0.041.

Read: Table 2:

- Column (1) estimates a separation effect of 0.110.
- Column (2) shows a similar effect, 0.121, for separation into non-employment.
- Column (3) implies 1.483 additional quarters of non-employment.
- Column (4) implies 0.971 additional quarters of unemployment benefit receipt.
- Column (5) implies 1.056 fewer quarters of continuous employment with the original employer.

Economic message: REBP does not merely reallocate workers to other jobs. It destroys matches and raises non-employment, creating the missing mass of marginal matches needed for the resilience test.



(a) Initial treatment effect, raw cohort evidence

TABLE 2

Initial treatment effect: difference-in-differences effects on separations (1988–93) among pre-reform job holders

	(1) Separation	(2) Separation into non-employment	(3) Non-employment (Quarters)	(4) Unemp. (Benefits) (Quarters)	(5) Cont. En (Quarte
REBP region × Treated cohort	0.110***	0.121***	1.483***	0.971*	−1.056*
	(0.041)	(0.043)	(0.380)	(0.532)	(0.370)
REBP region	0.003	−0.003	−0.235	−0.098	0.014
	(0.044)	(0.008)	(0.276)	(0.182)	(0.677)
Treated cohort	0.030	0.111***	0.817***	0.158***	0.154
	(0.026)	(0.004)	(0.129)	(0.058)	(0.392)
Constant	0.375***	0.058***	1.500**	0.682	15.956*
	(0.100)	(0.017)	(0.664)	(0.450)	(1.846)
Observations	384,200	384,200	384,200	384,200	384,20
Adjusted R^2	0.008	0.047	0.023	0.019	0.002
No of clusters	100	100	100	100	100

Notes: This table reports results of the econometric specification in equation (9). The coefficient of interest is that on REI Region × Treated Cohort, which captures the effect of REBP eligibility on the outcomes listed in Columns (1) throu (5) on a sample of workers employed at the onset of the reform (1988q2). We exclude workers born before 1933 and aft 1948. *Separation* denotes an indicator function that is 1 if a worker separated from their 1988-employer by the end of t REBP period (1988q2 to 1993q3). *Separation into non-employment* denotes an indicator for *Separation* from the initi employer interacted with an indicator for not taking up employment with another employer. *Non-employment (Quarter. Unemployment (Benefits) (Quarters)*, and *Continuous employment (Quarters)* denote the quarters of non-employme unemployment benefits, and continuous employment with the initial employer between 1988q2 and 1993q3. Standa errors clustered at the administrative region level are reported in parentheses. Levels of significance: * 10%, ** 5%, a *** 1%.

(b) Initial treatment effect, DiD estimates

6.4 Figure 5 and Table 3: post-repeal resilience test

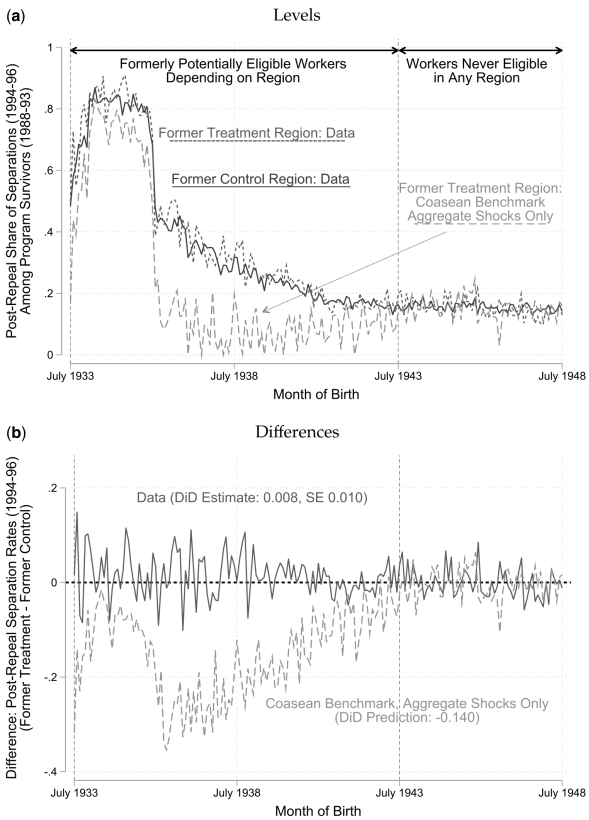
Read: Figure 5:

- The actual former treatment and former control survivor groups have very similar post-repeal separation rates.
- The actual DiD estimate is 0.008 with standard error 0.010.
- The Coasean benchmark with aggregate shocks predicts a much lower treatment-group separation rate.
- The benchmark DiD prediction is about −0.140, much more negative than the observed estimate.

Read: Table 3:

- The separation coefficient is 0.008 and statistically insignificant.
- Separation into non-employment is also 0.008 and insignificant.
- Non-employment quarters rise by 0.041, also insignificant.
- Unemployment benefit quarters fall by 0.071, insignificant.
- Continuous employment quarters fall by 0.079 and are significant, but small relative to the Coasean prediction.

Economic message: The key puzzle appears here. The policy destroyed marginal jobs, but after repeal the survivor group behaves as if it was not selected to be more robust.



(a) Post-repeal separations among survivors

TABLE 3

Resilience test: difference-in-differences effects on post-repeal separations (1994–6) among program survivors

	(1) Separation	(2) Separation into non-employment	(3) Non-employment (Quarters)	(4) Unemp. (Benefits) (Quarters)	(5) Cont. E (Quart
REBP region × Treated cohort	0.008	0.008	0.041	−0.071	−0.079
	(0.010)	(0.009)	(0.027)	(0.045)	(0.036
REBP region	−0.003	0.008	−0.007	0.006	0.108
	(0.019)	(0.011)	(0.056)	(0.042)	(0.091
Treated cohort	0.135***	0.160***	0.715***	0.149**	−0.581
	(0.008)	(0.003)	(0.008)	(0.071)	(0.048
Constant	0.152***	0.063**	0.306**	0.139	8.203
	(0.052)	(0.030)	(0.144)	(0.108)	(0.249
Observations	201,409	201,409	201,409	201,409	201,4
Adjusted R^2	0.025	0.046	0.038	0.006	0.016
No of clusters	99	99	99	99	99

Notes: This table reports results of the specification in equation (9). Here, the sample is restricted to workers employ at the same establishment in May 1988 and February 1994 i.e. survivors. The coefficient of interest is REBP Reg × Treated Cohort and captures the effect of REBP-eligibility on the outcomes listed in Columns (1) through (5), w outcomes measured by February 1996. We exclude workers born before 1933 and after 1948. *Separation* denotes indicator function that is 1 if a worker is not employed by their employer from February 1994 (and May 1988) in Febru 1996. *Separation into Non-employment* denotes an indicator for *Separation* from the initial employer interacted with indicator for not being employed in February 1996. *Non-employment (Quarters)*, *Unemployment (Benefits) (Quarte* and *Continuous employment (Quarters)* denote the quarters of Non-employment, unemployment benefits, and continu employment with the initial employer between February 1994 and 1996. Standard errors clustered at the administrat region level are reported in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

(b) Post-repeal DiD estimates

6.5 Figure 6 and Table 4: negative shocks and Coasean rationalizations

Read: Figure 6:

- Panel (a) splits the post-repeal cohort gap by industry growth terciles.
- Panels (b)–(d) relate separations to establishment employment growth.
- Separations rise when establishments shrink.
- Former treatment survivors are not more insulated from shrinking-establishment shocks.

Read: Table 4:

- Panel A estimates the share κ of cells that would need full reshuffling to rationalize the data.
- Estimates of κ are close to or above one across horizons and cohort definitions.
- Panel B allows idiosyncratic shocks; estimates still imply near-unit weight on full reshuffling.
- The estimates imply that a Coasean rationalization requires almost complete and rapid surplus reshuffling.

Economic message: Negative-shock responses reinforce the non-resilience result. Table 4 shows that Coasean rescue stories require extreme assumptions, not modest idiosyncratic noise.

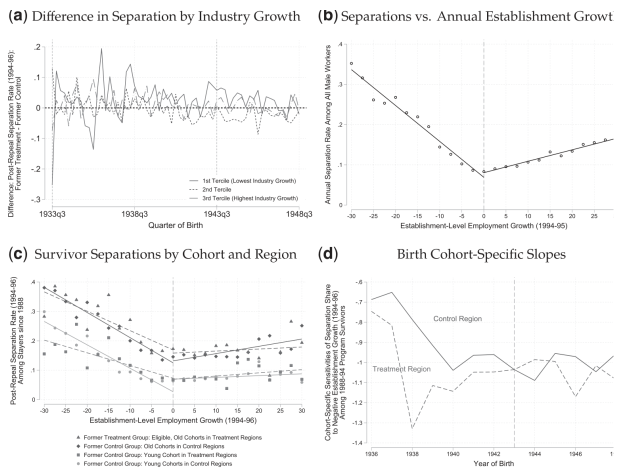


FIGURE 6

Resilience tests: post-repeal separation responses to negative industry and establishment-level growth events (1994–

Notes: (a) The by-cohort regional difference from Figure 5(b) split into terciles of industry growth, with the first tercile denoting the lov and the third tercile denoting the highest industry growth. Specifically, we calculate employment growth between 1994q1 and 199 for each industry (two-digit NACE), among all workers (not just stayers) born after 1938. Panels (b), (c), and (d) plot the results of analysis focusing on labour demand shifts within establishments. We confirm the "hockey-stick" relationship between separations employment growth at the establishment level (Davis, Faberman and Haltiwanger, 2013) in (b). It plots annual separation rates for n workers employed in a given year by bins of 1994q1–95q1 establishment employment growth. (c) The four REBP groups' (eligible ineligible cohorts and regions) separations against *total* establishment employment growth. We ignore the cohorts born before 1936 si

(a) Post-repeal response to negative shocks

TABLE 4

Mixed model estimates of share of cells with full post-repeal surplus reshuffling

	5-year cohorts				1-year cohorts			
Separation horizon:	1995 (1)	1996 (2)	1997 (3)	1998 (4)	1995 (5)	1996 (6)	1997 (7)	1998 (8)
Panel A: No post-repeal idiosyncratic shocks								
κ	0.978	1.040	1.097	1.117	0.974	1.083	1.157	1.22
	(0.035)	(0.044)	(0.057)	(0.077)	(0.032)	(0.035)	(0.043)	(0.05
$1-\kappa$	0.022	−0.040	−0.097	−0.117	0.026	−0.083	−0.157	−0.22
	(0.035)	(0.044)	(0.057)	(0.077)	(0.032)	(0.035)	(0.043)	(0.05
R^2	0.792	0.910	0.935	0.952	0.803	0.883	0.915	0.93
N	182	182	182	182	513	513	513	513
Panel B: Idiosyncratic shocks								
κ	0.934	1.045	1.097	0.889	0.864	0.992	1.157	0.93
	(0.053)	(0.052)	(0.057)	(0.096)	(0.051)	(0.008)	(0.043)	(0.04
$1-\kappa$	0.066	−0.045	−0.097	0.111	0.136	0.008	−0.157	0.06
	(0.053)	(0.052)	(0.057)	(0.096)	(0.051)	(0.008)	(0.043)	(0.04
$x_{1933-1938}$	0.080	0.265	0.001	0.233				
	(0.089)	(0.210)		(0.220)				
$x_{1938-1943}$	0.039	0.000	0.000	0.263				
R^2	0.793	0.909	0.935	0.950	0.809	0.881	0.915	0.92
N	182	182	182	182	513	513	513	513

Notes: This table reports NLS estimates of equation (12) (Panel A) and equation (15) (Panel B), assessing what fracti of (size-weighted) labour market cells would need to exhibit full post-repeal surplus reshuffling in order to rationali our empirical control and treatment group separation rates. Panel A estimates the simple specification described

(b) Mixed-model estimates

6.6 Figure 7 and Figure 8: calibrated benchmarks and wage rigidity

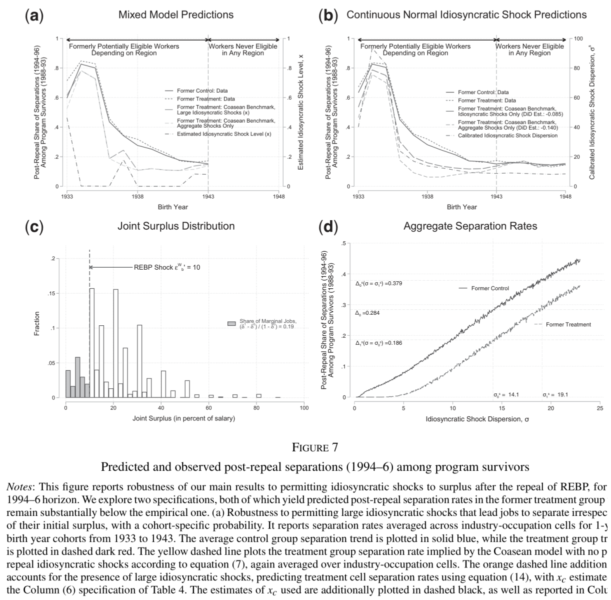
Read: Figure 7:

- Panels (a) and (b) compare actual post-repeal separations with Coasean predictions under mixed and continuous-shock benchmarks.
- The predicted treatment-group separation rates are below the empirical treatment-group separation rates.
- Panel (c) uses survey evidence on joint surplus distributions to calibrate the treatment survivor distribution.
- Panel (d) shows how aggregate separation rates vary with idiosyncratic shock dispersion.

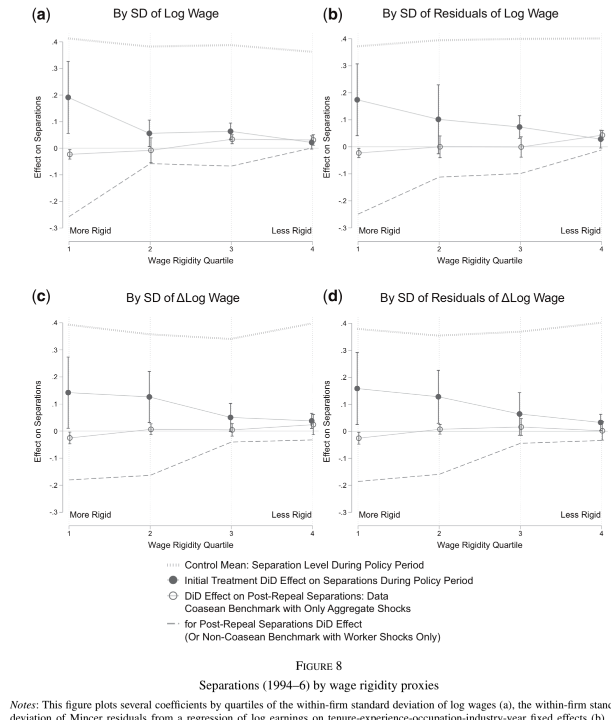
Read: Figure 8:

- The figure groups firms by wage-rigidity proxies, with more rigid cells on the left.
- The initial treatment effect is largest in rigid cells.
- The post-repeal data still show little resilience.
- The Coasean benchmark predicts substantial negative post-repeal effects especially in rigid cells, but the data do not show them.

Economic message: The final mechanism evidence points toward wage rigidity. Rigid wage cells are where REBP is most likely to destroy matches inefficiently and where the Coasean benchmark is most clearly rejected.



(a) Predicted and observed post-repeal separations



(b) Wage rigidity proxies

7 Detailed section-by-section notes

7.1 Section 1: Introduction

- The paper starts from the observation that bilateral efficiency is a benchmark assumption in labor economics. If worker and firm can bargain freely, no positive joint-surplus job should dissolve.
- The authors explain why this assumption is empirically elusive: surplus is not observed, and observed labels like layoff or quit do not reveal whether a match had positive joint surplus.
- The paper's key idea is to use a temporary policy that first extracts marginal matches and is then removed. The composition of survivors becomes informative about whether the removed matches were truly lowest joint-surplus matches.
- The introduction previews two facts: REBP strongly increased separations during the program, but former treatment survivors displayed no post-repeal resilience.
- The authors motivate wage rigidity as the leading explanation, while acknowledging that the wage-rigidity evidence is suggestive rather than definitive.

Incremental contribution: The introduction frames the paper as a revealed-preference test of an efficiency assumption that is otherwise hard to test directly. The novelty is not measuring surplus, but inferring surplus ranking from dynamic separation behavior.

7.2 Section 2: Institutional context, policy variation, and data

- The section describes the Austrian UI system and the Regional Extended Benefit Program.

- REBP is large enough to move separations: a jump from around half a year of UI to four years of UI creates a major outside-option shock for older workers.
- The treatment is not nationwide. Treated regions are compared with control regions, and older cohorts are compared with younger cohorts.
- The authors emphasize institutional features that matter for interpretation: quitters could receive UI after a waiting period, the reform did not alter payroll taxes or experience rating, and layoffs also faced severance and notice institutions.
- The data allow the authors to observe employer identity, employment spells, earnings, region, cohort, and benefit receipt.

Interpretation: Section 2 establishes the credibility of the empirical design. The policy shock must be large, clean, and temporary; the data must track exact matches through time.

7.3 Section 3: Deriving the test of the Coasean model

- The section formalizes the Coasean model and derives the resilience prediction.
- Under flexible bargaining, a match survives if joint surplus is non-negative. This makes the initial REBP-induced separations a way of identifying low joint-surplus matches.
- The key prediction is not simply that UI raises separations. It is that, after repeal, the former treatment survivor group should be missing low joint-surplus jobs.
- The benchmark equation maps observed control separations and during-program treatment effects into a predicted treatment survivor separation rate.
- The wage-rigidity model is introduced to show how non-resilience can arise: separations depend on unilateral worker and firm participation constraints, not just joint surplus.

Interpretation: This section is the theoretical core. It produces a falsifiable prediction from Coasean bargaining and an alternative model that can explain rejection.

7.4 Section 4: Initial treatment effect

- The section documents that REBP raised separations among jobs held at the onset of the reform.
- The visual cohort design shows a clean gap opening among cohorts eligible for REBP, while younger ineligible cohorts show no treated-control gap.
- The regression evidence quantifies the effect at about 11 percentage points.
- The effect is concentrated in separation into non-employment rather than job-to-job mobility.
- This initial treatment effect is important because it creates the missing mass of marginal jobs.

Interpretation: Section 4 validates the first step of the design. The policy did in fact extract a large set of matches, giving power to the later resilience test.

7.5 Section 5: Post-repeal resilience

- The section restricts to 1988 jobs that survived until the 1993 repeal.
- The Coasean model predicts that former treatment survivors should separate less after repeal.
- The raw data and regression estimates show no such resilience.

- The authors compare the data to the Coasean benchmark and find a large gap: the benchmark predicts strongly negative treatment-control differences, while the observed difference is near zero.
- The section also studies negative industry and establishment shocks, where the pecking-order logic should be especially strong. Former treatment survivors still are not more resilient.

Interpretation: Section 5 is the main empirical rejection of the Coasean view. The policy sorted the population, but not in the way efficient joint-surplus bargaining predicts.

7.6 Section 6: Alternative Coasean rationalizations

- The authors ask whether post-repeal idiosyncratic shocks might replenish the missing mass of marginal jobs in the treatment group.
- A mixed model estimates how much of the economy would need immediate full reshuffling. The estimates imply almost all cells would need to behave as if completely reshuffled.
- A survey-based surplus calibration also fails to generate enough separations in the former treatment group.
- The section considers heterogeneous treatment valuation, surplus homogeneity, large-firm substitution, severance pay, and other concerns.
- These rationalizations do not overturn the main result.

Interpretation: Section 6 strengthens the argument by showing that the rejection is not an artifact of the simplest Coasean benchmark. The data require extreme Coasean assumptions.

7.7 Section 7: Wage rigidity as a resolution

- The section examines wage rigidity proxies based on within-firm wage dispersion and residual wage dispersion.
- The idea is that firms with less wage differentiation may be less able to adjust wages for older workers whose outside options changed.
- The initial REBP separation effect is larger in more rigid cells.
- The post-repeal non-resilience pattern also aligns with rigidity: the Coasean benchmark predicts strong resilience where rigidity proxies suggest marginal jobs were extracted, but the observed data do not show it.
- The authors are careful that the rigidity proxies are not randomly assigned, so the evidence is suggestive.

Interpretation: Section 7 provides the mechanism narrative. It connects the empirical rejection of Coasean separations to wage-setting institutions that prevent efficient transfers between worker and firm.

7.8 Section 8: Conclusion

- The paper concludes that the Coasean view of efficient separations is rejected in this setting.
- The evidence suggests that wage rigidity drives inefficient separations.
- The authors highlight open questions: the ultimate source of wage rigidity, the causal status of rigidity proxies, and external validity beyond the REBP compliers.

Interpretation: The conclusion is deliberately measured. The paper rejects a broad Coasean prediction for the jobs affected by REBP, but does not claim all separations in all settings are inefficient.

8 Short summary

- The paper tests whether job separations are bilaterally efficient.
- It uses Austria's REBP, a large temporary UI benefit extension, as a shock to workers' outside options.
- REBP raises separations during 1988–93 by about 11 percentage points.
- Under a Coasean view, the jobs destroyed by REBP should be low joint-surplus jobs.
- After REBP repeal, former treatment survivors should therefore be more resilient to future shocks.
- The data show no resilience: former treatment and former control survivors separate at nearly identical rates.
- This non-resilience holds both on average and in response to negative industry and establishment shocks.
- Coasean rationalizations require almost complete post-repeal surplus reshuffling, which the authors view as implausible.
- A wage-rigidity model explains why REBP can destroy jobs that are low in worker surplus but not necessarily low in joint surplus.
- Wage-rigidity proxies provide suggestive mechanism evidence.
- The paper's incremental contribution is a revealed-preference design for testing the efficiency of separations without observing surplus directly.

9 Conclusion

9.1 Core conclusion

Jäger, Schoefer, and Zweimüller provide a revealed-preference test of Coasean efficient separations. The test uses the fact that a temporary outside-option shock should remove marginal low-surplus matches from the treated group. If bargaining is efficient, those treated survivors should later be more resilient than control survivors. They are not. This rejects the Coasean benchmark in the paper's setting.

9.2 Economic intuition

The intuition is that efficient bargaining makes total joint surplus the only separation-relevant state variable. REBP should remove the lowest joint-surplus jobs. Once the policy disappears, the treated survivor group should therefore be "stronger." But if wages are rigid, REBP can remove jobs with low worker surplus even if the firm side still has substantial surplus. Then the survivor group need not be resilient to future firm-side shocks.

9.3 Takeaway

The paper changes how to interpret separations induced by labor-market policies. A separation caused by a higher outside option need not be an efficient destruction of a low-surplus match. It may be an inefficient separation caused by wage-setting constraints. This has implications for UI, wage rigidity, monopsony, rent sharing, and macro models that assume efficient job destruction.